Disclosure Day

Compact proof that we most likely inhabit an OPH simulation

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OPH makes a falsifiable compression claim: five axioms plus two closure constants generate the observer-facing structure of our Universe, including many measured quantities, without using those measurements as inputs. A single lucky hit would mean little. The strength of the argument is overconstraint: one frozen branch must simultaneously produce quantum readout, the Standard Model gauge quotient and its integers, the source-side fine-structure root, the charged-family shape, the QCD-free weak hierarchy, and the geometric gravity normalization—and the same integer triple and the same two closure constants must reappear across those sectors with no per-sector adjustment. Any one match could be luck; the claim is the conjunction. Independent witnesses may multiply only after shared ancestry and search multiplicity are charged, so each accepted row must carry a source and dependence audit. The source side is small: where the standard description spends roughly thirty tuned inputs, OPH carries two computed endpoints and a small integer package. Two numbers do not have room to memorize that much independent structure; if the structure appears anyway, it was generated, not stored. **The headline coincidence illustration: multiplying the deliberately coincidence-friendly window fractions itemized in Section 5 gives roughly 1 in 10^9—and about 1 in 10^3 after inflating every remaining window a further 100-fold in coincidence's favor.** This is not a posterior probability that OPH is correct or incorrect. Section 5 states the preregistered compression protocol needed to turn the illustration into a formal test.

Once the rules and readout maps are fixed before comparison, the finite branch has one output. OPH is wrong if that output leaks targets, admits an equally valid rival branch, misses the measured structures it claims to generate, or fails the stated tests. Numbers like $\alpha(0)$, SI G, hadronic inputs, PDG tables, cosmology scorecards, hardware lanes, and bridge-defined capacities remain checks and failure points until OPH derives them without using the measured answer.

What this proof contributes

The proof gives one readable path through OPH. Companion papers carry the longer derivations. Standard tools keep their usual meanings. OPH adds finite observer-patch closure, $P_{\mathrm{src}} = P_{\mathrm{cand}}$, $N_\star$, and the rule that a row counts only when the declared map emits it. That small source has to carry every sector in this note, and its failure tests, in one argument.

Anti-fitting rule

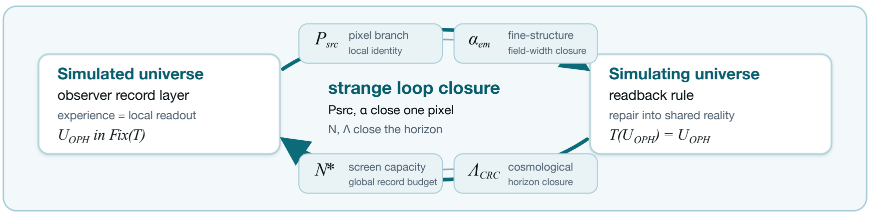
A numerical row is counted in our argument only if the paper or code gives four things: the calculation that emits it, no hidden path from the measured answer into that calculation, an interval or uniqueness check, and a clear way for the row to fail. A measurement may identify the basin to inspect. The endpoint must come from the source map. If the target value, or an algebraically equivalent calibrated proxy, appears in the calculation path, the row is comparison data, a display check, empirical closure, or a test surface. It sits outside the compact anti-circularity proof.

Most people picture a simulation as a machine that renders a world frame by frame. OPH uses a different mechanism: a fixed-point computation without a global timeline.

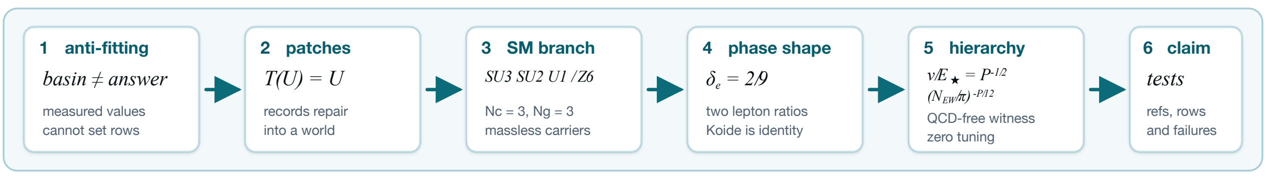
$$\text{ordinary simulation: } U_t \to U_{t+1} \to U_{t+2} \to \cdots, \qquad \text{OPH: } \mathcal{T}(W) = W.$$

The symbol W is the globally self-consistent observer-readable structure. $\mathcal{T}$ is the OPH readback-and-repair operator over observer patches, records, overlaps, mismatch scores, repair moves, and checkpoint continuation [O1,O2,O3]. The universe is the normal form of that computation. From inside, observers read it as history and experience that reading as time. Hofstadter's name for this kind of thing is a strange loop: a "paradoxical level-crossing feedback loop" [S16]. In OPH, strange loops sit at the core of the finite observer construction. Every row below carries one of three evidence classes: internal theorem, retrodictive source-only match, or frozen prospective prediction; anything that reads the measured answer is demoted to comparison data. The counted rows follow the anti-fitting rule above.

Bracketed source keys matter. When this note names a theorem without printing the full proof, the numbered OPH paper carries the proof details. The direct reading anchors at the end point to the online source lines. For fast lookup: OPH axioms are [A1], consensus repair is [A2], pixel closure is [A4], capacity closure is [A5], the QCD-free hierarchy witness is [A10], and the Standard Model gauge quotient is [A21]. The derivation chains printed here and in the referenced papers are stated so that each step can be checked end-to-end; if you find an error, contact info@floatingpragma.io.



The labels name what the values do: P_{src} fixes the local pixel branch used in the compact witnesses, α is the field-width readback, $N_\star$ fixes total screen capacity, and Λ is the horizon closure seen in cosmology.



OPH's five axioms are listed in *Observers Are All You Need* [O1]. Think of them as the toolchain. The argument below uses the finite observer-patch normal form, the exact quantum event surface, the source pixel coordinate, the global capacity target, the recovered Standard Model branch, and one compact three-scalar cross-lock:

$$(N_c, N_g, \beta_{\rm EW}) = (3, 3, 4) \quad \Longrightarrow \quad \left(\frac{v}{E_\star}, \frac{m_\mu}{m_e}, \frac{m_\tau}{m_e}\right).$$

The table says which rows are counted in this compact argument and which rows stay comparison data or falsification targets [O1,O3,O4,O5,O7,O8,O9].

1. The fixed-point computation

In OPH, "simulation" has a stricter meaning than "fixed point." The theorem-level test is the OPH simulation test [O2,A23]. It requires five independent properties: endogenous update, observer-readable records, overlap repair with a schedule-independent normal form, selected-branch elimination, and implementation/clock closure. The mathematical object is a finite patch system. Patches expose ports, write records, compare overlaps, repair disagreement, and return a shared normal form after readback [O1,O2,O3]. Internal observers experience that normal form as the universe.

The symbol $\mathfrak{U}_{\rm OPH}$ denotes the selected OPH universe. The map T is the observer-overlap repair map [O1,O2]. The notation $\mathrm{Fix}(T)$ means the states returned unchanged by that map:

$$T(\mathfrak{U}_{\rm OPH}) = \mathfrak{U}_{\rm OPH}.$$

The fixed-point equation follows from the five-clause simulation test. The proof in [O2,A23] first checks those clauses on the declared finite branch and then derives the identity. Ordinary equilibria fail this test; the same reference gives simple counterexamples and lists the failure modes. This note cites that proof instead of repeating it.

Conditional OPH closure theorem

Fix OPH axioms, branch rules, and quotient/readout maps before test data, and require source maps to be forced by those rules up to observer-invisible equivalence. Let

$$\Gamma_P : I_P \to I_P, \qquad \Gamma_N : I_N \to I_N$$

be source-only contractions. Suppose their dependency graphs contain no held-out target path, no admissible source-map deformation survives the axioms and branch-selection rules, and the finite packet passes the simulation test of [O2,A23]. Then

$$P_\star = \Gamma_P(P_\star), \qquad N_\star = \Gamma_N(N_\star)$$

have unique solutions. Every declared quotient observable

$$Y_i = \pi_i(\mathrm{nf}_{\rm OPH}(P_\star, N_\star))$$

is uniquely determined and invariant under admissible implementation, scheduling, and hidden-carrier changes. The hard part is showing that OPH leaves no rival source map; Banach then supplies endpoint uniqueness. The theorem proves unique generation from the frozen branch. Physical identification requires the data comparison.

Read the equation literally. The simulated record world enters the simulating readback rule, contradictory branches die, and the returned object is the same record world. That is the self-contained consistency computation [O1,O2]. A universe with unexplained external constants fails this test [O1,O3,O4]. The finite OPH patch-carrier pipeline is the concrete fixed-cutoff carrier for this computation: a finite patch system with ports, central records, repair moves, checkpoints, and a terminal normal form. It carries the closure solve; frame rendering is absent. Two verification layers pin the consensus/repair mathematics down. The reference benchmark suite replays all six overlap schedules and reports zero schedule divergence of the repaired normal form on the clean abelian branches, with a designed nonabelian stale-record counterexample as negative control [C5]. A standalone Lean 4/Mathlib artifact machine-checks the abstract layer—observer confluence, approximate schedule independence, refinement naturality, and the contractive conditional-resampling projector—in 98 sorry-free theorems; the OPH-specific consensus theorem remains an open formalization obligation [C6].

The observer lock changes the meaning of experience. An observer in OPH is a finite record boundary with a comparison rule for nearby patches [O1,O2]. Subjective experience is that boundary reading itself from the inside:

$$r_i = \mathcal{R}_i(\mathfrak{U}_{\rm OPH}), \qquad \mathfrak{U}_{\rm OPH} = T(\{r_i\}_{\rm compat}).$$

In this equation, r_i is observer patch i's local readout, $\mathcal{R}_i$ is its readout map, and $\{r_i\}_{\rm compat}$ is the compatible readout set. The map T repairs those records and returns the shared world. Left equation: subjective experience. Right equation: shared reality. Experience matters because OPH reality is the stable object made by compatible observer readouts [O1,O2]. Private incompatible records fail to become public records. The mechanism is mundane: records read, compare, repair, return.

Exact quantum event-surface hit

The fixed-cutoff record interface also emits textbook quantum readout. On a completed compare/write/verify slice, observer-accessible events generate a finite central record algebra. For an event E,

$$\mathbb{P}(E) = \mathrm{Tr}(\rho P_E), \qquad \rho|_E = \frac{P_E \rho P_E}{\mathrm{Tr}(\rho P_E)}.$$

These are the Born rule and the Lüders update on the OPH record surface. On the corresponding two-wing Bell event surface, the CHSH expression obeys

$$|S_{\rm CHSH}| \le 2\sqrt{2}.$$

This is an exact structural output of the finite record algebra. It uses no $\alpha(0)$, G, QCD, charged-lepton masses, or neutrino data [O1,O3,A18,A19]. A finite 65,536-record patch run reproduces this surface numerically: committed-event Born weights sum to 1.0 at machine precision, the record projectors commute with zero idempotency defect, and Lüders re-reads are stable with repeat-read fraction 1.0 [C7].

2. Closure coordinates and counted witnesses

Standard physics carries roughly thirty independent numerical inputs, depending on how the count is made. OPH leaves a local pixel coordinate and a global screen-capacity coordinate. They are fixed by closure equations from the OPH rules [O1,O3,O4,O5,O6]. That is the zero-parameter claim; branch localization follows the anti-fitting rule.

For every counted compact calculation, write

$$P_{\mathrm{src}} := P_{\mathrm{cand}} = 1.630972095694329\ldots.$$

This is the source-side pixel candidate emitted by the declared OPH trunk. The CODATA/NIST comparison coordinate is

$$P_C := \varphi + \frac{\sqrt{\pi}}{A_{\mathrm{C}}}, \qquad A_{\mathrm{C}} := 137.035999177 = 1.630968209403959\ldots.$$

The CODATA/NIST measured endpoint is a branch locator and comparison column only; it supplies no upstream source-map input. The global screen target remains

$$N_\star \simeq N_{\mathrm{scr}}^{\mathrm{dS}} \simeq 3.31 \times 10^{122}.$$

In ordinary language: one elementary observer cell must read the same from the simulated record side and the simulating screen side, and the universe must carry the record capacity required by the screen that contains that same universe.

Independent two-closure lock

The OPH universe is a self-reference. The simulating side supplies a readback rule, the simulated side supplies the record readout, and a consistent universe exists only when both sides close on the same object [O1,O2,O4]. The two closure equations are where this strange loop becomes a number. **The closure coordinates must be computed as OPH endpoints. The simulation is consistent only when the self-references hold.**

Local loop: try a pixel P_k. The OPH branch reads its field width $R_P(P_k)$. The pixel map turns that readout back into a corrected pixel:

$$\alpha_k = R_P(P_k), \qquad P_{k+1} = \Gamma_P(P_k) = \varphi + \alpha_k\sqrt{\pi}.$$

Solving $P = \Gamma_P(P)$ gives the simulating-side pixel setting for that branch. In the compact hierarchy row this coordinate is $P_{\mathrm{src}} = P_{\mathrm{cand}}$. The simulated-side field width comes from the same endpoint: $\alpha_{\mathrm{root}} = R_P(P_{\mathrm{src}})$.

Global loop: try a capacity N_k. The capacity map counts closed-screen normal forms, charges the screen budget, applies the Minimal Admissible Realization (MAR) selector, and returns the admissible capacity:

$$\Lambda_k = R_N(N_k), \qquad N_{k+1} = F_N(N_k).$$

Solving $N = F_N(N)$ gives the simulating-side capacity setting $N_\star$. The simulated-side horizon readout comes from the same computed endpoint: $\Lambda_\star = R_N(N_\star)$.

Solvers may start in observation-suggested intervals. The equations themselves must contain no measured α, measured Λ, weak masses, calibrated target residuals, or equally admissible deformations. Once the source maps are fixed and contractive, Banach gives one local endpoint and one global endpoint:

$$P_{\mathrm{src}} = \Gamma_P(P_{\mathrm{src}}), \qquad N_\star = F_N(N_\star).$$

Changing either endpoint breaks the resonance between the source setting and the universe read from inside [O1,O3,O4,O5].

The OPH screen branch supplies twelve curvature ports; reversible write/check orientation doubles the product-adjoint repair register to 24. Both counts reappear as the exponent and the normalization of the hierarchy proof below. This is where the fixed-point simulation stops being a metaphor.

The endpoints have a limited carrier reading: P_{src} fixes a canonical cell area, $N_\star$ fixes total capacity, and the equal-area cell count is $K_{\mathrm{cell}} = 4N_\star/P_{\mathrm{src}}$; the primitive observer algebra itself stays open unless a separate theorem selects a primitive carrier.

The local constant is computed from a readback equation with measured α excluded [O3,O4,O5,O6]. The symbol P is a candidate identity value for one OPH pixel: the number that converts the cell's screen area into the edge entropy it shares with neighboring cells. One pixel supports two readings. The simulating side reads the geometric width $\alpha_{\mathrm{ext}}(P) = (P-\varphi)/\sqrt{\pi}$: the golden ratio $\varphi = (1+\sqrt{5})/2$ is the balance point of a pixel with no field readback, $\sqrt{\pi}$ converts boundary width into the pixel chart, and the electromagnetic width supplies the detuning. The simulated side reads the field width through the OPH branch chain

$$P \longmapsto \alpha_U(P) \longmapsto A_Z(P) \longmapsto A_T^{\mathrm{src}}(P), \qquad \alpha_{\mathrm{in}}(P) = \frac{1}{A_T^{\mathrm{src}}(P)},$$

where $\alpha_U(P)$, $A_Z(P)$, and $A_T^{\mathrm{src}}(P)$ are the finite-screen unified gauge width, the electroweak anchor, and the Ward-projected Thomson endpoint. The pixel lock is their shared value:

$$P_{\mathrm{src}} = \Gamma_{\mathrm{src}}(P_{\mathrm{src}}) = \varphi + \frac{\sqrt{\pi}}{A_T^{\mathrm{src}}(P_{\mathrm{src}})}, \qquad \alpha_{\mathrm{root}} = \alpha_{\mathrm{ext}}(P_{\mathrm{src}}) = \alpha_{\mathrm{in}}(P_{\mathrm{src}}).$$

The public implementation [C1] solves the same loop in the α-chart: a trial α_k defines $P_k = \varphi + \alpha_k\sqrt{\pi}$, the branch returns $\alpha_{\mathrm{in}}(P_k)$, and bisection drives the mismatch $\rho(\alpha_k) = \alpha_{\mathrm{in}}(P_k) - \alpha_k$ to zero inside the declared branch interval. The output is the pair

$$P_{\mathrm{src}} = P_{\mathrm{cand}}, \qquad \alpha_{\mathrm{root}}^{-1} = 136.9948351646\ldots.$$

Measured α may identify the branch interval; it never defines A_T^{src} or Γ_{src}, and CODATA can only be printed after the solve as a comparison. Reading the lock backward with the measured A_{C} gives the comparison coordinate P_C above; the code path in [C1] keeps comparison values out of the solver.

Fine-structure prediction and QCD endpoint

The fine-structure row is the sharp numerical stress test in this compact proof. With no measured $\alpha(0)$, no Thomson endpoint, and no hadron data in the input path, the trunk emits $\alpha_{\mathrm{root}}^{-1} = 136.9948351646\ldots$, reproducible from the public solver and checked runtime report [C1]:

```
cd code/P_derivation && python3 derive_p.py --no-hadron-closure
```

The saved report `runtime/full_p_alpha_report_current.json` prints $\alpha^{-1} = 136.994835164621649\ldots$; the measured CODATA value never enters that command path. Adding the unified width $\alpha_U(P_C) = 0.041124336195630495$, evaluated at the CODATA-derived comparison pixel, gives the mixed-provenance no-hadron diagnostic

$$A^{\rm fp}_{\alpha_U} = \alpha^{-1}_{\rm root} + \alpha_U(P_C) = 137.0359595008\ldots, \qquad A_{\rm C} - A^{\rm fp}_{\alpha_U} = 3.96762 \times 10^{-5},$$

a relative gap of 2.8953×10^{-7} against the measured $A_{\rm C} = \alpha^{-1}(0) = 137.035999177(21)$. This is CODATA-minus-diagnostic bookkeeping, not a source-only prediction, and the missing source object sits exactly where the trunk stops: low-energy electromagnetic dressing by confined quark and hadron spectral transport. In transport form, $A_{\rm C} - \alpha^{-1}_{\rm root} = \alpha_U(P_C)\, C_{24,Q}$ with $C_{24,Q} = 1.00096478597\ldots$; a 24-register spectral object fitted after looking at the Thomson residual would be exact accounting rather than prediction. Source-only closure of this row requires the Ward-projected hadronic spectral measure and same-scheme finite endpoint remainder without Thomson-target leakage—a finite-lattice QCD computation outside this compact proof; OMEGA is the optical fixed-point compute lane for such payloads [S14]. Until that map is emitted, CODATA/NIST $\alpha(0)$, hadron masses, and the SI G clock row stay in the calibrated comparison lane [O5,O6], $P_{\rm full}$ names the hadron-completed endpoint, and the compact count uses the QCD-free hierarchy witness below, which avoids the hadronic correction completely.

The global constant is computed from the screen-capacity readback equation [O1,O4]:

$$N_\star = \mathcal{C}(N_\star).$$

The symbol N is a candidate total capacity for the closed observer screen: the number of record units the universe can carry in OPH units. The simulated reading counts the closed-screen normal forms supported at capacity N; the simulating reading charges the screen for the N slots it claims. A self-contained fixed-point computation has no external memory bank, so the available records and the cost of the screen must close on the same capacity [O1,O4]. The screen solver mirrors the pixel solver: try a capacity N, count the closed-screen normal forms it supports, subtract the screen budget charge, and apply MAR to pick the admissible branch. The score is

$$S(N) = \log|\Omega^{\rm sc}_N| - N, \qquad N_\star = \mathrm{MAR}\arg\max_N S(N) \simeq 3.31 \times 10^{122}.$$

The set $\Omega^{\rm sc}_N$ contains the closed-screen normal forms at capacity N. The operator MAR is Axiom 5 of OPH [O1], Minimal Admissible Realization: the realized branch is the lexicographically minimal admissible representative under the branch complexity vector $C(\mathfrak{S})$; here it selects the minimal admissible arithmetic representative of the maximizing branch. The cosmological constant comes from that same capacity:

$$\Lambda_\star \ell_\star^2 = \frac{3\pi}{N_\star}, \qquad \Lambda_\star a_{\rm cell} = \frac{3\pi P_{\rm src}}{N_\star}.$$

So $N_\star$ and $\Lambda_\star$ are computed together in the global closure, the same way $P_{\rm src}$ and $\alpha_{\rm root}$ are computed together in the local closure [O1,O4].

The three-scalar cross-lock

Before any decimal appears, the recovered compact-gauge branch fixes the Standard Model quotient

$$G_{\rm SM} = \frac{\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y}{\mathbb{Z}_6}, \qquad N_c = 3, \qquad N_g = 3.$$

The derivation has a specific route. Combined zero-obstruction transport—central or higher-associator strictification plus an allowed trivial-holonomy strict representative—first identifies individually transportable seeds. One common strict representative is then fixed at the stage, and only seeds trivial under that same choice generate its tensor category. On a cofinal tail carrying the explicit compact-gauge refinement receipt, surjective pullback functors and forgetful fibers build the refinement-limit bosonic category, and DR/Tannaka reconstruction reads off a compact group from it. MAR then selects the realized one-generation/one-Higgs matter package; anomaly cancellation, Yukawa invariance, and the central kernel of the realized action fix the hypercharge lattice and the $\mathbb{Z}_6$ quotient [O4,A21]. The same branch carries the protected massless photon, gluon, and graviton rows:

$$m_\gamma = 0, \qquad m_g = 0, \qquad m_{\rm grav} = 0.$$

The same quotient fixes charge quantization: $Q = T_3 + Y$ and $Q_{\rm color\ singlet} \in \mathbb{Z}$. Its connected adjoint is $(8,1,0) \oplus (1,3,0) \oplus (1,1,0)$. There are no mixed $(3,2,\pm 5/6)$ X/Y gauge bosons, so the ordinary simple-GUT gauge-mediated proton-decay channel is absent on this branch [O4,A20,A21]. This discrete structural witness sits upstream of the numerical rows. The same branch then emits the compact integer triple

$$N_c = 3, \qquad N_g = 3, \qquad \beta_{\rm EW} = N_c + 1 = 4.$$

These three integers feed both compact numerical witnesses below. The same 4 controls the charged-family phase and the weak hierarchy exponent [O3,O4,O5]. One source branch gives one huge hierarchy and a two-coordinate lepton-family shape.

Phase-rigidity charged-family prediction. The charged-family row is one correlated family-shape prediction: no QCD, no G, no public $\alpha(0)$, and no charged-lepton masses enter as inputs. On the realized Standard Model branch,

$$N_c = 3, \qquad N_g = 3, \qquad \beta_{\rm EW} = N_c + 1 = 4.$$

Koide's relation, $(\sum_i m_i)/(\sum_i \sqrt{m_i})^2 = 2/3$, has been an unexplained empirical regularity of the charged leptons since 1981. In OPH it stops being a mystery: every balanced three-family carrier satisfies it as an exact trace identity, and the OPH-specific claim is the phase selector that picks the realized direction on that cone from branch integers:

$$\delta_e = \frac{\beta_{\rm EW}}{2N_cN_g} = \frac{2}{9}.$$

If R is the cyclic three-family shift, $R^3 = I$, define

$$C_{\delta_e} = I + \frac{1}{\sqrt{2}}\left(e^{i\delta_e}R + e^{-i\delta_e}R^2\right).$$

Its eigenvalues are

$$r_k = 1 + \sqrt{2}\cos\left(\delta_e + \frac{2\pi k}{3}\right), \qquad k = 0, 1, 2.$$

For $\delta_e = 2/9$, the ordered roots are

$$(r_e, r_\mu, r_\tau) = (0.040349908219207475,\ 0.5802119201475368,\ 2.3794381716332555).$$

The trace identities are immediate:

$$\operatorname{tr} C_{\delta_e} = 3, \qquad \operatorname{tr} C_{\delta_e}^2 = 6.$$

If charged square-root masses lie on this OPH carrier direction, $m_i = s_e r_i^2$, then Koide follows exactly:

$$\frac{\sum_i m_i}{(\sum_i \sqrt{m_i})^2} = \frac{s_e \operatorname{tr} C_{\delta_e}^2}{s_e (\operatorname{tr} C_{\delta_e})^2} = \frac{6}{9} = \frac{2}{3}.$$

The phase-fixed direction predicts the two ordered scale-free charged-family ratios

$$\frac{m_\mu}{m_e} = \left(\frac{r_\mu}{r_e}\right)^2 = 206.770315972729\ldots,$$
$$\frac{m_\tau}{m_e} = \left(\frac{r_\tau}{r_e}\right)^2 = 3477.47283710460\ldots,$$
$$\frac{m_\tau}{m_\mu} = \left(\frac{r_\tau}{r_\mu}\right)^2 = 16.8180467333776\ldots.$$

Using the PDG lepton summary masses $m_e = 0.51099895000\,\mathrm{MeV}$, $m_\mu = 105.6583755\,\mathrm{MeV}$, and $m_\tau = 1776.86\,\mathrm{MeV}$, the deviations are about 9.8 ppm, 70 ppm, and 60 ppm respectively [S8]. The Koide value computed from those same pole masses is about 9.2 ppm below 2/3. The compact row counts one phase-rigidity prediction: $\delta_e = 2/9$ fixes the normalized charged-family direction, and the two ordered ratios test that direction jointly. This row fails if N_c, N_g, or β_{EW} lack an independent Standard Model-branch derivation, if $\delta_e = 2/9$ is chosen from charged-lepton masses, or if the charged-family shape misses the sorted lepton direction [O5,O6,A15,A16].

The QCD-free hierarchy witness

The hierarchy problem becomes a QCD-free branch-output calculation. A route through the public Thomson endpoint, hadron masses, or the cesium clock would import the Ward-projected hadronic correction as an assumption; this second independent witness avoids it entirely. The OPH calculation fixes $\alpha_U(P_{\mathrm{src}}) = 0.0411243336195630495$ in the interval $I_U = \alpha_U(P_{\mathrm{src}}) \pm 10^{-6}$, with the Krawczyk image strictly inside I_U [O5]. The counted row is the weak-to-UV hierarchy emitted from P_{src} and that independently checked $\alpha_U(P_{\mathrm{src}})$ value [O5,C3]. OPH gets the tiny weak scale from the two closure constants, without importing hadronic physics.

The factor 12 is screen topology doing the counting. On the triangulated S^2 screen, the locally flat triangular vacuum is six-valent. Define the curvature charge $q_v = 6 - \deg(v)$. Since $V - E + F = 2$ and $3F = 2E$, one has $E = 3V - 6$, hence

$$\sum_v q_v = 6V - \sum_v \deg(v) = 6V - 2E = 12.$$

A closed triangular screen must carry exactly twelve units of curvature. Refinement-stable MaxEnt spreads this total charge into twelve unit fivefold defects: at fixed $\sum q_v = 12$, the convex local defect cost $\sum q_v^2$ is minimized by unit charges, while negative defects require extra positive charge. Edge-center completion turns each defect collar into a central port [O3]. With no marked point, the twelve equal ports sit on the maximal finite rotation orbit A_5/C_5, whose size is $60/5 = 12$. Therefore the global screen depth $D = \log(N/\pi)$ is sampled locally as $D/12$.

The representation-to-spectrum step turns the same screen count into the oriented repair register:

$$m_{\mathrm{rep}} = 2\dim(\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)) = 2(8 + 3 + 1) = 24.$$

The factor two is reversible write/check orientation. Since $N = \pi R_{\mathrm{dS}}^2/\ell_\star^2$, the same $D = \log(N/\pi)$ is the cosmic length ratio in logarithmic form. The electroweak branch samples that global depth through the 12 unoriented ports and the $4P_{\mathrm{src}}$ projection below; the 24-slot register supplies the repair normalization [O4,O5,C3].

The local cell entropy is $\ell_{\mathrm{cell}} = P_{\mathrm{src}}/4$, and the electroweak projection coefficient is $\beta_{\mathrm{EW}} = N_c + 1 = 4$. Therefore

$$\Gamma_{\mathrm{EW}} = \beta_{\mathrm{EW}} \ell_{\mathrm{cell}} \frac{\log(N/\pi)}{12} = \frac{P_{\mathrm{src}}}{12} \log(N/\pi).$$

This is the local share of the global screen step: cell entropy, times the electroweak projection, times one twelfth of the screen depth. The compact standalone hierarchy witness is the transmutation law

$$\frac{v}{E_\star} = P_{\mathrm{src}}^{-1/2} \exp\left[-\frac{2\pi}{4\alpha_U(P_{\mathrm{src}})}\right] = 2.0199803239725553 \times 10^{-17}.$$

$E_\star$ is the OPH UV energy scale. $E_{\mathrm{cell}} = E_\star/\sqrt{P_{\mathrm{src}}}$ is the cell energy after the pixel area is accounted for. The exponent uses the OPH-emitted $\alpha_U(P_{\mathrm{src}})$. It excludes G, Λ, M_W, M_Z, Higgs mass, the public Thomson endpoint, and QCD hadronic input [O5]. This is the QCD-free quantitative witness.

The bridge coordinate rewrites the same exponent as a screen-capacity readout:

$$\frac{v}{E_{\mathrm{cell}}} = \left(\frac{N_{\mathrm{CRC}}^{\mathrm{EW}}}{\pi}\right)^{-P_{\mathrm{src}}/12}, \qquad N_{\mathrm{CRC}}^{\mathrm{EW}} = \pi \exp\left[\frac{6\pi}{P_{\mathrm{src}}\alpha_U(P_{\mathrm{src}})}\right] = 3.5323546226929906\ldots \times 10^{122}.$$

The affine map $\mathcal{C}_{\mathrm{EW}}(P,x) = (1-\lambda)x + \lambda\, 6\pi/(P\alpha_U(P))$, $\lambda = \frac{1}{2}$, is at fixed $P = P_{\mathrm{src}}$ a contraction whose unique endpoint is $x_\star = \log(N^{\mathrm{EW}}_{\mathrm{CRC}}/\pi)$; the residual $\alpha_U(P_{\mathrm{src}})^{-1} - (P_{\mathrm{src}}/6\pi)\log(N^{\mathrm{EW}}_{\mathrm{CRC}}/\pi)$ vanishes there by construction, and the finite readback-resolution result supplies the positive-root limit $\rho_{\mathrm{read}}(r, N^{\mathrm{EW}}_{\mathrm{CRC}}) \to (N^{\mathrm{EW}}_{\mathrm{CRC}}/\pi)^{-1/2}$ [O5,C3]. This bridge capacity is bookkeeping, not a counted row—but it lands next to the global closure target: the screen depths $\log(N^{\mathrm{EW}}_{\mathrm{CRC}}/\pi) = 281.03$ and $\log(N_\star/\pi) \simeq 280.97$ agree to 2.1×10^{-4}. On this branch the weak hierarchy and the horizon capacity are one magnitude statement; the remaining $\sim 7\%$ linear capacity gap is a named target for the source capacity row, not a counted match. A source-only global capacity row carries empirical weight only after an independently motivated executable map computes screen capacity without measured Λ, horizon radius, age, or N_{scr}, and then compares the frozen output with cosmology [O4,O5,C3].

Ordinary particle physics treats the Higgs scale as a tuning problem. In OPH the weak scale is a branch output with zero naturality defect:

$$\epsilon_H = 0, \qquad \epsilon_H \in [0,0].$$

Zero means there is no cancellation to arrange. There is no bare scalar dial to balance against a cutoff; the weak scale is the settled-form readout of the 24-slot oriented repair-register normalization. That is the hierarchy problem in standard language, with the standard fine-tuning term absent from the OPH normal form [O4,O5].

Discrete rigidity check

The integers $N_c = 3$, $N_g = 3$, and $\beta_{\mathrm{EW}} = 4$ are branch data with no extra measured decimals. The evidence is the overconstraint. Once the Standard Model branch fixes this integer triple, the same β_{EW} hits two unrelated readouts: the weak hierarchy and the charged-family shape. Hold the P_{src} and $\alpha_U(P_{\mathrm{src}})$ proof records fixed and change only β_{EW}:

β_{EW}	$v/E_\star$	m_μ/m_e	m_τ/m_e
3	5.97×10^{-23}	25.84	579.07
4	2.01998×10^{-17}	206.7703	3477.47
5	4.20×10^{-14}	outside cone	outside cone

The $\beta_{\mathrm{EW}} = 5$ charged carrier has

$$r_e = 1 + \sqrt{2}\cos\left(\frac{5}{18} + \frac{2\pi}{3}\right) = -0.0158500583\ldots,$$

so the light square-root mass leaves the positive cone. The nearby branch values miss both sectors at once. The 4 is therefore a positive-edge branch: it keeps the hierarchy in the observed sector and produces a tiny positive electron root without fitting charged masses [O4,O5,C3].

Prediction escrow: rows exposed to failure

Known constants can point to branch neighborhoods. The case needs frozen outputs outside the charged-family and hierarchy rows, exposed to external measurements.

Charged-family residual. The known Koide relation is background. The OPH claim is the branch phase and the residual. Define

$$\eta_e := \delta_e^{\mathrm{phys}} - \frac{2}{9}.$$

The compact row uses $\delta_e = 2/9$ as a branch cross-lock with the hierarchy integer triple. A full charged-family promotion has to emit η_e and the affine determinant-line anchor from the same OPH branch [O5,O6,A15,A16].

Running-scheme rigidity. The hierarchy row counts only when the $\alpha_U(P_{\mathrm{src}})$ proof record fixes the high-scale beta packet, threshold packet, representation cutoffs, matching prescription, and renormalization convention outside the target value. Tuning any of those with $v/E_\star$ as the target demotes the row to calibration. A stronger closure emits the running packet from the same OPH branch, or emits a threshold spectrum that experiments can exclude [O5,C3,A8,A10].

3. Geometric gravity normalization

Einstein theorem boundary

Bare finite consensus supplies quotient normal forms; Einstein dynamics belongs to the declared branch $\mathsf{Cons}_r + \mathsf{GeomRead}_r + \mathsf{BW}_r + \mathsf{NullStress}_r + \mathsf{BoundedInterval}_r + \mathsf{FixedCapStat}_r + \mathsf{SmallBallArea}_r + \mathsf{TensorUpgrade}_r$, with D6 capacity closure for Λ. Failure of any branch clause leaves the gravity row unclosed.

OPH gets gravity before meters, kilograms, or seconds enter the discussion. G_{geom} is Newton coupling in OPH geometric units, a_{cell} is one observer-cell area, $\bar{\ell}_{\mathrm{shared}}$ is its shared edge-entropy weight, and $\ell_\star$ is the OPH length quantum. The edge-entropy/area-law theorem on the OPH gravity branch identifies $a_{\mathrm{cell}}/(4\bar{\ell}_{\mathrm{shared}})$ with the structural coefficient of the stress-energy term in the local Einstein equation $G_{ab} + \Lambda g_{ab} = 8\pi G_{\mathrm{geom}}\langle T_{ab}\rangle$ and in the Newton-Poisson weak-field limit $\nabla^2\Phi = 4\pi G_{\mathrm{geom}}\rho$; the same G_{geom} propagates literally through the Jacobson chain into both limits [O4,A22]. On the realized local-pixel branch that theorem reduces to

$$G_{\mathrm{geom}} = \frac{a_{\mathrm{cell}}}{4\bar{\ell}_{\mathrm{shared}}}, \qquad a_{\mathrm{cell}} = P_{\mathrm{src}}\ell_\star^2, \qquad \bar{\ell}_{\mathrm{shared}} = \frac{P_{\mathrm{src}}}{4}.$$

The A22 derivation fixes the factor 4. Substitution gives

$$\frac{G_{\mathrm{geom}}}{\ell_\star^2} = \frac{P_{\mathrm{src}}}{4(P_{\mathrm{src}}/4)} = 1, \qquad G_{\mathrm{geom}} = \ell_\star^2.$$

The same pixel number counts area and shared entropy, so the pixel number cancels: gravity is the area quantum in OPH units. The theorem uses no measured G, Planck area, measured Λ, or gravity-calibrated clock scale [O4,A22]: the linear-in-area scaling $\langle L_C\rangle \propto A(\partial C)$ is derived from the Markov-collar/refinement-stability structure of the screen net, the structural Einstein coefficient is $\kappa_{\mathrm{OPH}} = 2\pi\, a_{\mathrm{cell}}/\bar{\ell}_{\mathrm{shared}}$, and the Newton-Poisson convention $\nabla^2\Phi = 4\pi G_N\rho$ fixes $G_N = \kappa_{\mathrm{OPH}}/(8\pi) = a_{\mathrm{cell}}/(4\bar{\ell}_{\mathrm{shared}}) = G_{\mathrm{geom}}$; the factor 4 is the ratio $8\pi/(2\pi)$ between the two conventions. The SI decimal needs the separate no-G clock proof [O4,A6].

4. Compact proof rows

The compact argument separates three evidence classes. Internal theorems show mathematical coherence of the declared branch. Retrodictive matches compare source-only outputs with known quantities. Prospective predictions are frozen outputs exposed to later tests. The compact proof surface consists of: observer-patch architecture, the exact quantum event surface, the Standard Model structural branch, the phase-fixed charged-family shape, the QCD-free hierarchy witness, and the geometric gravity normalization identity. OPH claims to describe the whole machine, so the longer particle, cosmology, dark-sector, hardware, and SI rows matter. They stay outside this compact surface unless their own calculations pass the same rule [O1,O4,O5,C2,C3].

Rows outside the compact count remain useful evidence and falsification surfaces with their own claim classes. The readable falsifiability map is `extra/OPH_falsifiability.md`.

Witness	Evidence class	Why	Refs
Finite observer-patch normal form and patch-carrier pipeline	Internal theorem	Finite observer records, overlaps, mismatch scores, repair maps, checkpoints, and patch carriers define the fixed-point architecture. The row tests the mechanism before any decimal fit enters.	[O1,O2,C5]
Lean-verified consensus/repair layer	Internal theorem	A sorry-free Lean 4/Mathlib artifact (98 theorems, axiom-audited) machine-checks observer confluence, approximate schedule independence, refinement naturality, and the contractive conditional-resampling projector. The OPH-specific consensus theorem is an open formalization obligation.	[C6]
Born-Lüders and Tsirelson event surface	Internal theorem	The finite central record algebra gives $\mathbb{P}(E) = \mathrm{Tr}(\rho P_E)$, Lüders conditioning, and $\lvert S_{\mathrm{CHSH}}\rvert \leq 2\sqrt{2}$ on the declared Bell event surface; a 64k-record run reproduces it numerically.	[O1, O3, A18, A19, C7]
Standard Model structural branch	Internal theorem	$\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y/\mathbb{Z}_6$, $N_c = 3$, $N_g = 3$, protected massless carriers, charge quantization, and no X/Y bosons are discrete structural outputs feeding both numerical witnesses.	[O3, O4, O5, A20, A21]
$\alpha_U(P_{\mathrm{src}})$ interval check	Internal theorem	The unified-coupling row is checked by an interval/Krawczyk calculation with measured electroweak masses, low-energy couplings, G, Planck units, and Λ excluded.	[O5,C2,A8]
Phase-fixed charged-family shape	Retrodictive match	$\delta_e = (N_c + 1)/(2N_c N_g) = 2/9$ fixes the scale-free charged-family direction. Koide is an identity of the carrier; the two ordered lepton ratios test the single normalized direction. Absolute lepton masses stay outside this row.	[O5,O6,A15,A16]
QCD-free hierarchy law	Retrodictive match	$v/E_\star = P_{\mathrm{src}}^{-1/2} \exp[-2\pi/(4\alpha_U(P_{\mathrm{src}}))]$. The weak scale is emitted without the QCD/hadronic endpoint and $\epsilon_H = 0$, conditional on the independent meaning of $E_\star$.	[O5,C3,A10]
Geometric gravity normalization	Internal identity	$G_{\mathrm{geom}} = \ell_\star^2$. This fixes the OPH area normalization. The SI Newton decimal needs the separate clock row.	[O4,A7,A22]
CODATA/NIST $\alpha(0)$ and P_C	Comparison only	The measured endpoint requires the missing QCD/hadronic electromagnetic transport. $P_C = \varphi + \sqrt{\pi}/A_{\mathrm{C}}$ is a comparison coordinate under the declared missing-map condition.	[O5,O6,A13]
SI G	Comparison only	The SI decimal counts only when the no-G cesium clock hierarchy emits $\varepsilon_{\mathrm{Cs}}$ without measured G or gravity-calibrated proxies.	[O4,A6]
Particle, dark-sector, neutrino, cosmology, and collider rows	Prospective prediction	These rows remain useful tests with their declared claim classes. The compact anti-circularity argument keeps them outside its count.	[O5,O7,O8,C2,C4]
PIXEL, IBM, OMEGA, and hardware lanes	Experimental surface	Hardware lanes test named lemmas. The universe claim depends on the source rows and data comparison. They count only with public task definitions, raw readouts, controls, manifests, and verifier logs.	[S12,S13,S14]

The compact rows in this table are derived from OPH closure coordinates and the OPH axioms, with the full derivation chains and proofs contained in the referenced papers. Comparison and test rows keep their declared evidence class and failure conditions. For details, see the linked papers and the public code in `github.com/FloatingPragma/observer-patch-holography`.

5. Falsifiers and evidence classes

OPH has many ways to be wrong: source leakage, nonunique closure, a MAR rival, carrier-dependent readout, hidden-scale bridge, or missed frozen prediction. One failure demotes its row; a central failure kills the compact claim. The empirical test is a frozen compression score. Freeze the theory, source maps, branch selectors, tolerances, uncertainty model, comparison theories, and held-out dataset D_{test}. Then compare

$$\Delta_{\mathrm{bits}} = \log_2 \frac{p(D_{\mathrm{test}} \mid M_{\mathrm{OPH}})}{p(D_{\mathrm{test}} \mid M_{\mathrm{ref}})} - \big[L(M_{\mathrm{OPH}}) - L(M_{\mathrm{ref}})\big].$$

The length $L(M)$ charges the full description of the model: axioms, MAR, source equations, branch rules, selectors, tolerances, and search. The empirical identification claim is that a source-locked OPH generation compresses the preregistered held-out observations by at least the declared threshold Δ_0 bits more than the reference class. This is the joint-improbability argument in exact form: every unadjusted hit adds bits that a rival model must buy with fitted parameters, and $L(M)$ makes post-hoc machinery expensive by construction.

A ballpark coincidence budget

The multiplication can be illustrated with deliberately coincidence-friendly windows. Grant the accident hypothesis generous per-row odds: the one-phase charged-family direction landing both ordered lepton ratios at the observed ≤ 70 ppm ($\sim 10^{-4}$); the source-only fine-structure root landing within 3×10^{-4} of measured $\alpha^{-1}(0)$ with a residual of hadronic size and sign ($\sim 10^{-3}$); the exact Standard Model quotient with three colors and three generations ($\geq 10^{-2}$, charitably). The product is $\sim 10^{-9}$—about 30 bits—before counting the Born/Tsirelson structure, gravity normalization, SPARC, or Λ at all. In plain terms: **the assigned window product is roughly 1 in 10^9, or 0.0000001%. Inflate every remaining window a further 100-fold in coincidence's favor and that product becomes about 0.1% (1 in 10^3).** This is an illustration, not a posterior: the windows are judgment calls, independence is approximate, and selection over discarded branches would deflate it—which is exactly why the counted protocol freezes maps first and charges $L(M)$ for every rule. The illustration shows the scale that the frozen protocol is designed to certify.

Too many problems fall out at once

String theory earned four decades of attention largely because one structure falls out of it unasked: quantized strings contain a spin-2 graviton. In OPH the protected massless spin-2 branch falls out of the observer-overlap reconstruction in the same unasked way [O4,O9,S15]—and it is one entry in a list. One fitted constant is cheap; the claim a reader should price is the conjunction below, all emitted by the same two fixed points and one integer triple, with no per-problem machinery added.

Observer problem: experience is a finite readout boundary; public reality is the compatible fixed point of those readouts [O1,O2].

Gravity with gauge physics: the Einstein-branch geometry readout, Lorentz kinematics, the $H^3 \simeq \mathrm{SO}^+(3,1)/\mathrm{SO}(3)$ three-dimensional spatial chart, the Jacobson-Einstein branch, compact gauge reconstruction, the Standard Model quotient, hypercharge, three colors, and three generations sit in one observer-overlap reconstruction stack [O3,O4,O5].

Hierarchy problem: the 12-port screen sieve, the 24-slot oriented repair register, and the local/global capacity bridge emit the weak scale with $\epsilon_H = 0$; the usual fine-tuning term is absent [O4,O5,C3].

Koide relation: the 1981 charged-lepton coincidence $Q = 2/3$, unexplained in the Standard Model, becomes an exact trace identity of the branch carrier, with phase $\delta_e = 2/9$ fixed by $(N_c, N_g, \beta_{\mathrm{EW}})$ [O5,O6].

Yang-Mills gap: on a controlled compact-gauge branch, once the continuum proof in [O10] is supplied, $\Delta_{\mathrm{YM}} = \Delta_{\mathrm{rep}} > 0$; without it, finite repair remains regulator-level gap accounting [O4,O10].

Dark energy: the cosmological constant is the closed-screen capacity readout of the same Einstein branch; arbitrary vacuum-energy input has no role in that row [O1,O4].

Dark matter: repair-stress bookkeeping on the declared dark branch replaces the unseen-particle assumption. The static galaxy continuation gives $\nu_{\mathrm{OPH}}(x) = [1 - \exp(-\lambda_{\mathrm{collar}}\sqrt{x})]^{-1}$, with $g_{\mathrm{obs}} \simeq \sqrt{a_{\mathrm{eff}} g_b}$ and $v^4 = G M_b a_{\mathrm{eff}}$ in the deep low-acceleration limit [O7,O8,C4].

Unification without proton decay: the edge branch runs unification-style while the realized product quotient has no simple-GUT leptoquark generators [O4,O5].

Particle inventory: the compact proof counts structural carriers, gauge quotient data, the phase-fixed charged-family shape, the $\alpha_U(P)$ interval check, and the QCD-free hierarchy witness; W/Z, absolute charged-lepton masses, selected-frame quarks, Higgs/top, and neutrino rows keep their declared support levels [O4,O5,C2].

String vacuum selection: the observer-patch constraints define a vacuum sieve over critical-string candidates; the Bouchard-Donagi witness is selected only behind the critical-edge, cohomology, safety-layer, threshold, and moduli-locking gates. This is the precise sense in which string theory is an effective description inside the OPH stack [O9,S15].

Tests outside the compact proof count

The next entries are clean ways to break OPH branches. They sit outside the compact proof until their own calculation and checks are supplied without reading the measured answer.

Collider tests [O5,C2,S6]. The particle files carry concrete collider-facing surfaces, including $m_H = 125.1995\,\mathrm{GeV}$, $m_t = 172.3524\,\mathrm{GeV}$, $M_W = 80.377\,\mathrm{GeV}$, and $M_Z = 91.18798\,\mathrm{GeV}$ (full precision and scheme caveats in [O5,C2]), and no low-scale visible gauge branch incompatible with the recovered quotient [O4,O5]. Falsification: scheme-controlled measurements exclude the stated surfaces, or CERN/FCC finds a light visible gauge boson or chiral exotic incompatible with the branch.

Dark-sector cosmology [O7,C4]. The honest current row is galactic. On the public SPARC rotation-curve sample, with systematic profiling over mass-to-light, distance, inclination, and gas, the OPH unit branch scores $\chi^2/\mathrm{pt} = 1.017$ against 1.023 for the same-function empirical reference, with $11.25\,\mathrm{km\,s^{-1}}$ velocity RMS and 0.13 dex all-point radial-acceleration scatter [C4]. This is a pre-likelihood scorecard. Test: first emit a finite covariant collar-packet parent with stress closure, recipient stress and exchange-current closure for any nonzero exchange branch, gauge-independent B_A, active-fiber/physical-clock receipts for any promoted Γ_{rec}, refinement convergence, and frozen likelihood hashes; then run the public dark-sector scorecard in [C4] and replace the compact SPARC scaffold by the full covariance likelihoods named in [O7]. Falsification: the fixed branch fails the preregistered galaxy, cluster, or weak-lensing tests.

Neutrino closure [O5,C2,A17]. The earlier weighted-cycle candidate is rejected and demoted per the anti-fitting rule; coordinates and audit detail live in the falsifiability map. A future counted neutrino row requires a source-emitted operator, closed charged basis, physical mass-label rule, and pre-registered hash.

Appendix A. Assigning a number to Newton's constant

The geometric gravity row in the compact argument is

$$G_{\mathrm{geom}} = \ell_\star^2.$$

The SI decimal is a separate clock translation. It counts only when the cesium gap is emitted without measured G, Planck units, or gravity-calibrated proxies [O4,A6]:

$$\varepsilon_{\mathrm{Cs}}^{\mathrm{calc}} = \mathcal{H}_{\mathrm{Cs}}\left(\alpha, \frac{m_e}{E_\star}, \frac{\Lambda_{\mathrm{QCD}}}{E_\star}, \{y_q\}, \mathcal{N}_{133}, \mathcal{B}_{\mathrm{atom}}^{\mathrm{Cs}}\right).$$

If $\varepsilon_{\mathrm{Cs}}$ is back-solved from measured G, the row is a display check. With an independently emitted clock gap, exact SI constants c, $\hbar$, and $\nu_{\mathrm{Cs}} = 9\,192\,631\,770\ \mathrm{s}^{-1}$ give [S3,S4]

$$G_{\mathrm{SI}} = \frac{c^5}{4\pi^2\hbar\nu_{\mathrm{Cs}}^2}\varepsilon_{\mathrm{Cs}}^2 = 6.674299995910528\ldots \times 10^{-11}\ \mathrm{m}^3\,\mathrm{kg}^{-1}\,\mathrm{s}^{-2}.$$

The finite-lattice QCD part of the cesium calculation targets the OMEGA compute class [S14]; this appendix is orientation only and sits outside the compact anti-circularity proof.

References

Data and external references

[S1] CODATA value for Newton's constant G.

[S2] CODATA inverse fine-structure constant α^{-1}.

[S3] CODATA exact speed of light c.

[S4] CODATA reduced Planck constant $\hbar$.

[S6] Particle Data Group gauge and Higgs-boson summary tables.

[S7] Particle Data Group quark summary tables.

[S8] Particle Data Group lepton summary tables.

[S9] NuFIT 6.1 neutrino oscillation fit.

[S10] KATRIN direct neutrino-mass result.

[S11] Particle Data Group neutrino-mass review.

[S12] PIXEL optical verification plan.

[S13] Frozen IBM quantum-cloud engineering archive; not OPH-versus-QM evidence.

[S14] OMEGA optical fixed-point compute hardware.

[S15] The Right String Theory, OPH blog reference writeup.

[S16] Douglas R. Hofstadter, *I Am a Strange Loop*, Basic Books.

OPH papers

[O1] Observers Are All You Need.

[O2] Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics.

[O3] Federated Echosahedral Screen Microphysics: Patch Hardware, Records, and Observer Synchronization in OPH.

[O4] Recovering Relativity and the Standard Model from Observer Overlap Consistency.

[O5] Deriving the Particle Zoo from Observer Consistency.

[O6] Fine-Structure Constant Derivation.

[O7] Observer-Patch Holography and the Dark Matter Phenomenon.

[O8] Theoretical Bounds on χ_ν in Observer-Patch Holography.

[O9] Observer Patch Holography as String Vacuum Selector.

[O10] Yang-Mills Gap and the Clay Problem.

Code and supporting data

[C1] Pixel-constant derivation code.

[C2] Particle-branch code and supporting files.

[C3] Hierarchy checks and local/global resonance code.

[C4] Dark-sector scorecard and continuation code.

[C5] Consensus and fixed-point architecture code, including the six-schedule reference benchmark suite.

[C6] Lean 4/Mathlib machine-checked consensus, repair, and confluence artifact with build and axiom-audit receipts.

[C7] OPH physics simulation: 64k-record universe runs with Born/Lüders record receipts.

Direct reading anchors (theorem keys → source papers; each key is named where cited)

[A1] → [O1]; [A2], [A23] → [O2]; [A3], [A18] → [O3]; [A19] → [O1] synthesis fragments; [A4], [A13], [A15] → [O6]; [A5]–[A7], [A9], [A14], [A20]–[A22] → [O4]; [A8], [A10]–[A12], [A16], [A17] → [O5].